# Algorithm Visualization with TikZ

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# A biconnected plane graph with three faces

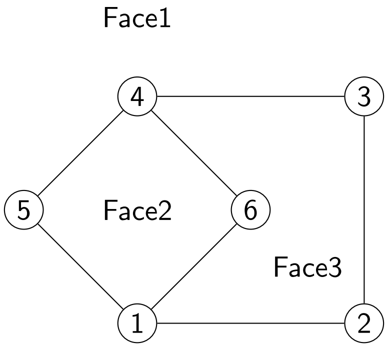
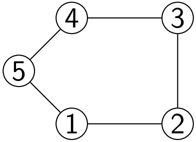
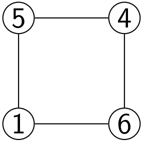



Figure: Root triangulation with safe root vertex 1. Chords from vertex 1 to all non-neighboring vertices (3, 4, 5) form a fan structure, ensuring no multi-edge conflicts with previously added chords.

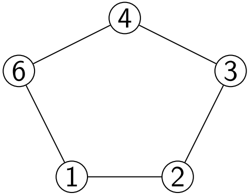
# Faces of the Graph




(a) Face 1

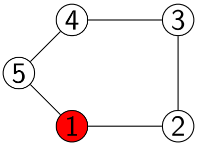



(b) Face 2

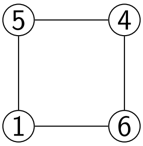



(c) Face 3

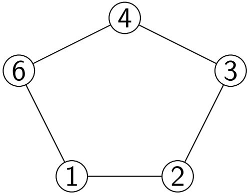
# Find safe root of face 1




(a) **Face 1**



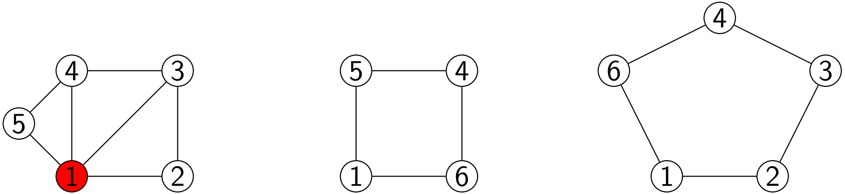

(b) Face 2




(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6)

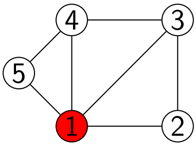
# Root Triangulation of face 1



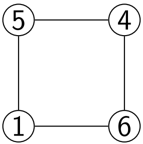

(a) **Face 1** (b) Face 2 (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

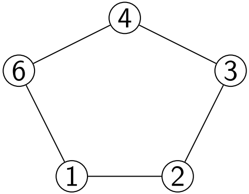
# Now we move to face 2




(a) Face 1



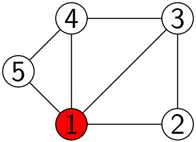

(b) **Face 2**



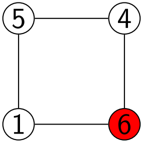

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

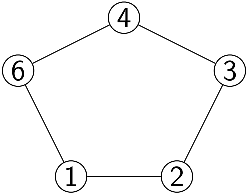
# Find safe root of face 2




(a) Face 1



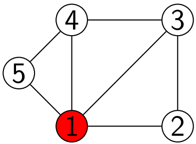

(b) **Face 2**



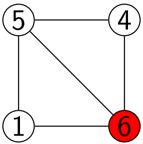

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

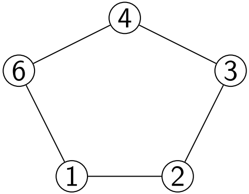
# Root Triangulation of face 2




(a) Face 1



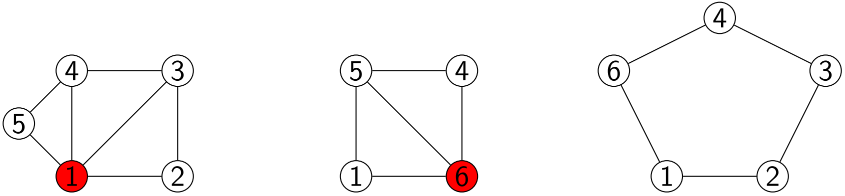

(b) **Face 2**




(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

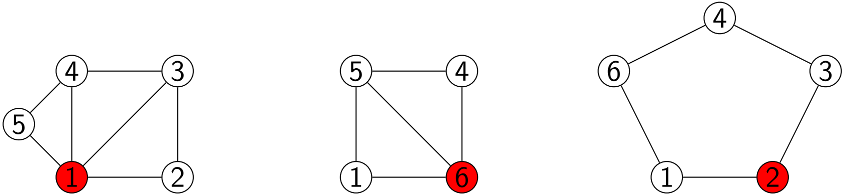
# Now we move to face 3




(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

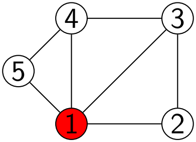
# Find safe root of face 3



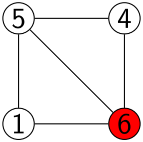

(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

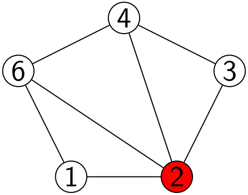
# Root Triangulation of face 3




(a) Face 1



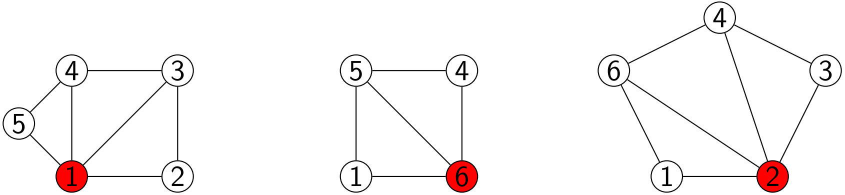

(b) Face 2




(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

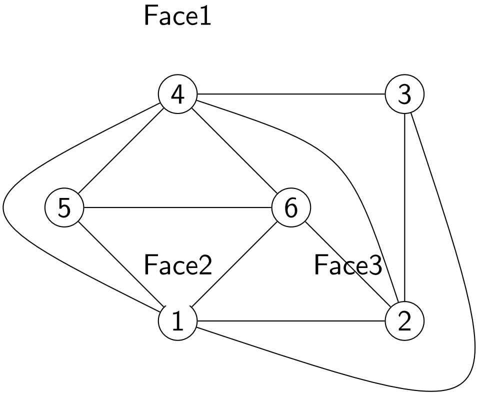
# There is no other faces after face 3, so its a complete valid triangulation




(a) Face 1 (b) Face 2 (c) **Face 3**

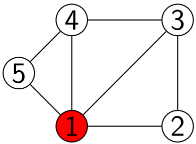
Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

# So the actual triangulation looks like this:

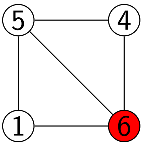



Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

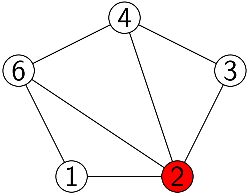
# Now we will start flipping of Face 3




(a) Face 1



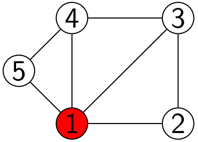

(b) Face 2



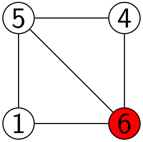

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

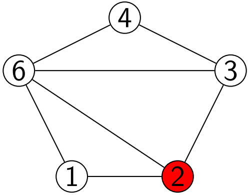
# We can flip the (2,4) edge, as its opposite pair is not in the global edge set




(a) Face 1



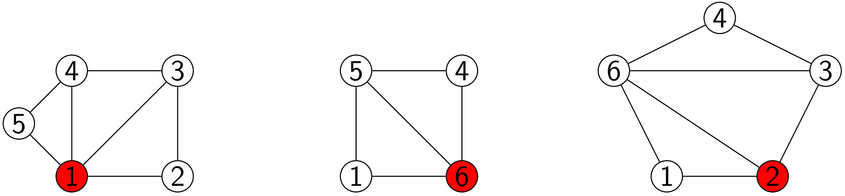

(b) Face 2




(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (3,6), (2,6)**

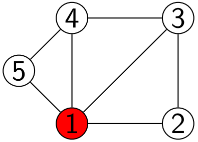
# Its a new triangulation



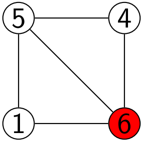

(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (3,6), (2,6)**

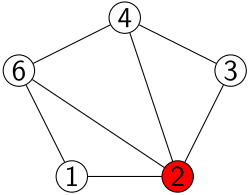
# in the root also, cant flip (2,6) as (1,4) is present in the global edge set




(a) Face 1



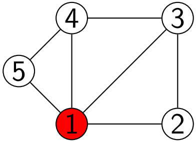

(b) Face 2



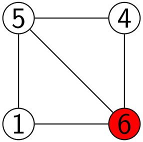

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

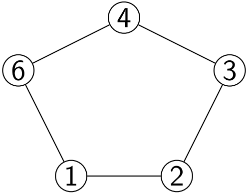
# So, there is no more triangulations possible from here, so we backtrack to face 2




(a) Face 1



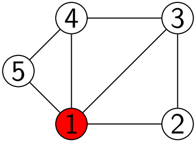

(b) **Face 2**



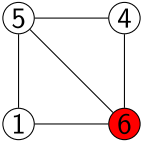

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

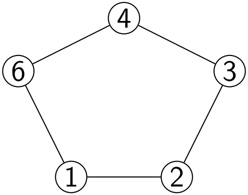
# We cant flip the (5,6) edge as (1,4) is present in the global edge set




(a) Face 1



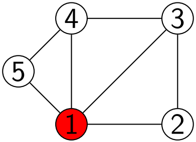

(b) **Face 2**



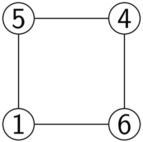

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

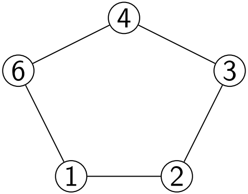
# So we backtrack to face 1




(a) **Face 1**



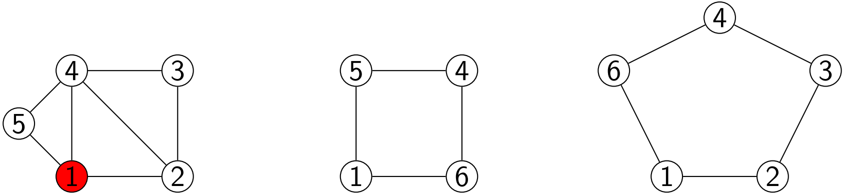

(b) Face 2




(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

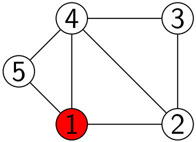
# We can flip the (1,3) edge, as its opposite pair is not in the global edge set



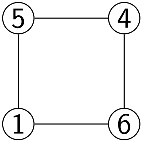

(a) **Face 1** (b) Face 2 (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

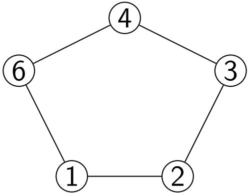
# Now we move to face 2




(a) Face 1



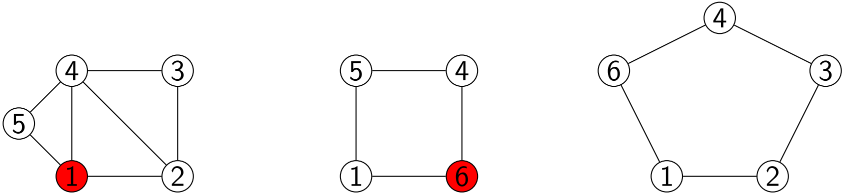

(b) **Face 2**




(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

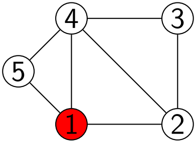
# Find the safe root of face 2



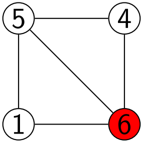

(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

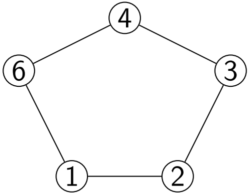
# Root Triangulation of face 2




(a) Face 1



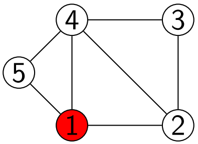

(b) **Face 2**



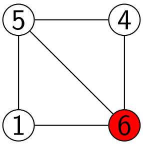

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4), (5,6)**

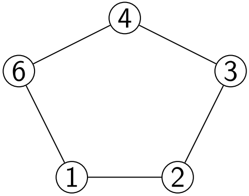
# Now we move to face 3




(a) Face 1



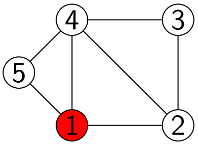

(b) Face 2



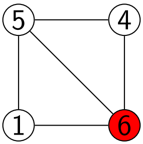

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4), (5,6)**

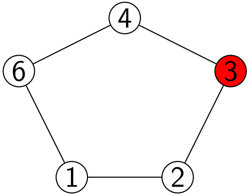
# Find safe root of face 3




(a) Face 1



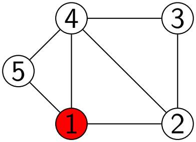

(b) Face 2



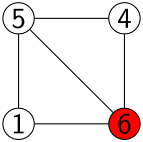

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4), (5,6)**

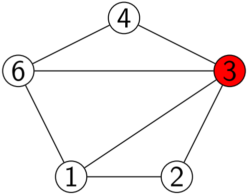
# Root Triangulation of face 3




(a) Face 1



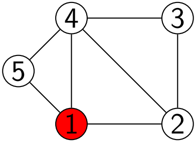

(b) Face 2



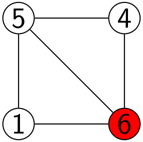

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (1,3), (3,6)**

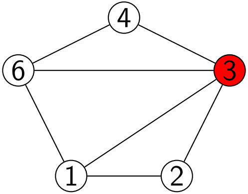
# There is no face after face 3, so its a complete triangulation




(a) Face 1



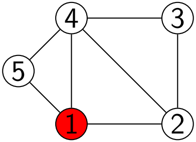

(b) Face 2



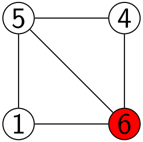

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (1,3), (3,6)**

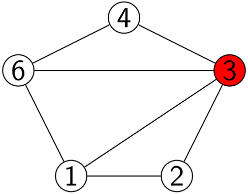
# Cant flip (3,6), because opposite pair is (1,4), already exists




(a) Face 1



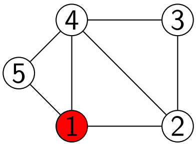

(b) Face 2



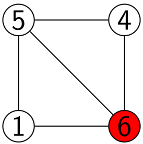

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (1,3), (3,6)**

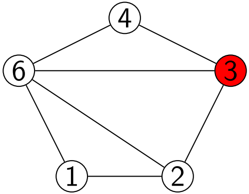
# we can flip (1,3) to (2,6), its a new triangulation




(a) Face 1



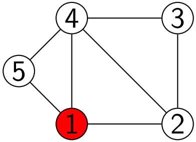

(b) Face 2



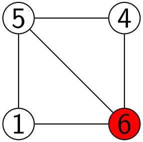

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (2,6), (3,6)**

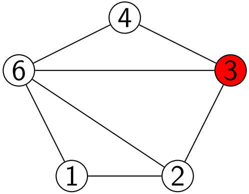
# Cant flip (3,6), because opposite pair is (2,4), already exists




(a) Face 1



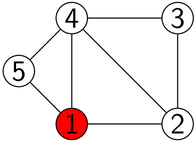

(b) Face 2



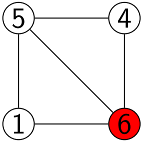

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (2,6), (3,6)**

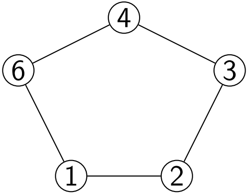
# So, there is no more triangulations possible from here, so we backtrack to face 2



(a) Face 1



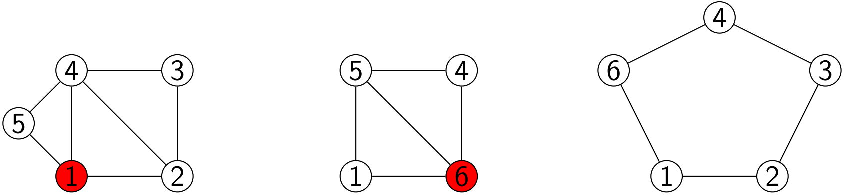
(b) **Face 2**



(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6)**

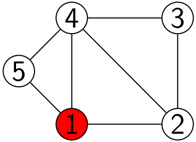
# We cant flip the (5,6) edge as (1,4) is present in the global edge set



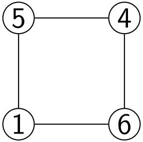

(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6)**

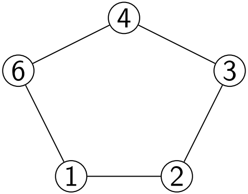
# So we backtrack to face 1




(a) **Face 1**



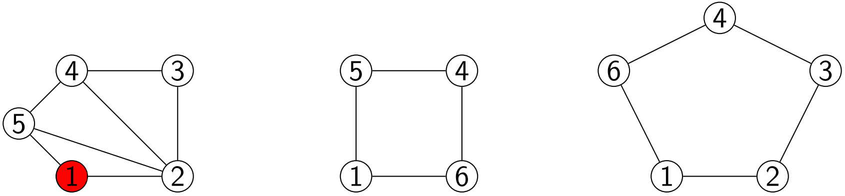

(b) Face 2




(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

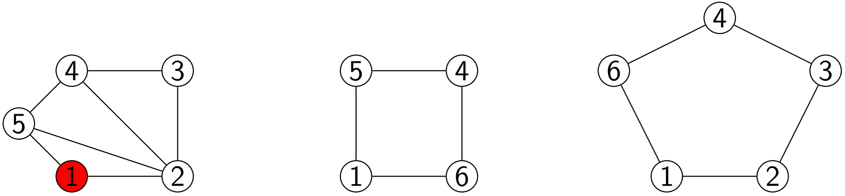
# We can flip (1,4) to (2,5)




(a) **Face 1** (b) Face 2 (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

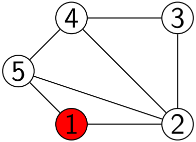
# We can now move to Face 2



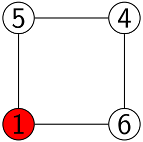

(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

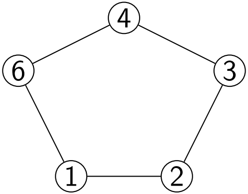
# Find safe root of Face 2




(a) Face 1



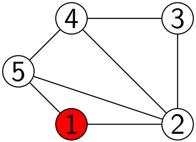

(b) **Face 2**



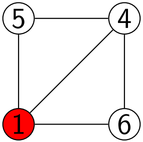

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

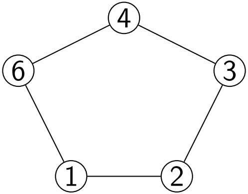
# Root Triangulation of Face 2




(a) Face 1



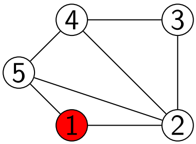

(b) **Face 2**



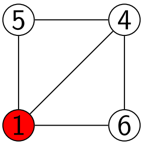

(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4)**

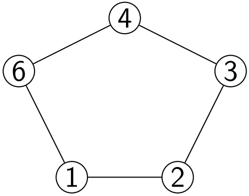
# Now we move to Face 3




(a) Face 1



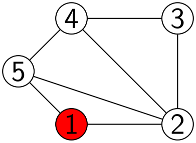

(b) Face 2



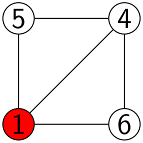

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4)**

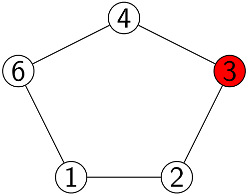
# Find safe root of Face 3




(a) Face 1



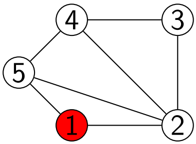

(b) Face 2



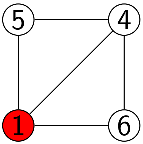

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4)**

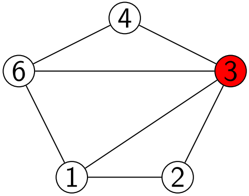
# Root Triangulation of Face 3



(a) Face 1



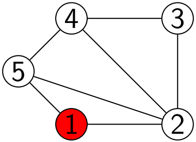
(b) Face 2



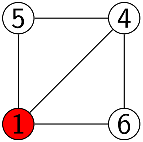
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (1,3)**

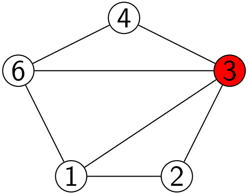
# There is no more faces, so its a complete triangulation




(a) Face 1



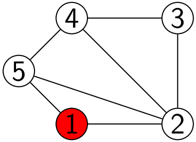

(b) Face 2



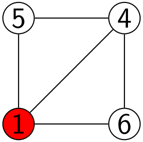

(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (1,3)**

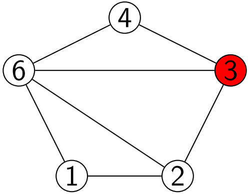
# We can flip the (1,3) edge to (2,6) to get a different triangulation




(a) Face 1



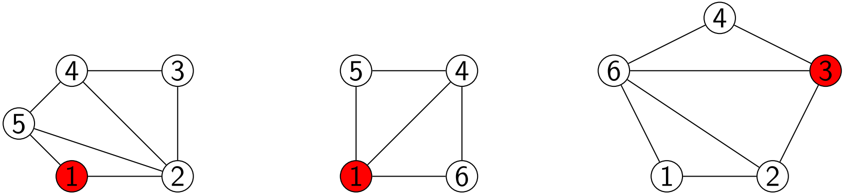

(b) Face 2




(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (2,6)**

# There is no more faces, so it's a complete triangulation




(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (2,6)**

# Continue....

We will be continueing this process until all faces are triangulated.